



AWS Architecture Practice Evaluation: Serverless Order Pipeline

Candidate: Golden Jacket Aspirant **Reviewer:** AWS Solutions Architect Coach **Project Focus:** Serverless Event-Driven Architecture (API Gateway -> Lambda -> SNS -> SQS -> Lambda -> DynamoDB)



Executive Summary

The candidate demonstrated exceptional architectural thinking, iterating through a rigorous 7-step active learning framework. The candidate moved beyond basic "happy path" tutorials and successfully engineered a highly decoupled, fault-tolerant, multi-queue architecture using Infrastructure as Code (CloudFormation).

Overall Coach Rating: Exceptional (Exam-Ready)



Step-by-Step Performance Review

Step 1: Draw Architecture by Hand

- **Candidate's Action:** Designed a fan-out architecture on paper.
- **Coach's Assessment: Outstanding.** The candidate correctly identified that the prompt implied a fan-out pattern and placed SNS *before* SQS. When challenged on missing compute layers and imprecise terminology (Queue vs. Topic), the candidate instantly iterated and produced a flawless v2 diagram clearly defining the SQS -> Lambda -> DynamoDB worker boundaries.

Step 2: Build it in Dev (Infrastructure as Code)

- **Candidate's Action:** Deployed the architecture using AWS CloudFormation.
- **Coach's Assessment: Advanced.** The candidate successfully navigated several complex CloudFormation integrations:
 - **Single-Table Design:** Correctly challenged if separate queues should feed a single DynamoDB table. Implemented the AWS Best Practice of Single-Table Design using `OrderType` tags.
 - **Smart Routing:** Caught a critical logical flaw where orders would duplicate to the wrong queue. Fixed this by implementing SNS `FilterPolicy` configurations.
 - **Visualizer Limitations:** Encountered and diagnosed a CloudFormation limitation where inline subscriptions do not support `FilterPolicy`, successfully refactoring to standalone resources.

Step 3: Break One Thing Intentionally

- **Candidate's Action:** Injected a "poison pill" (`"vehicle": "FAIL_CAR"`) into the system.
- **Coach's Assessment: Spot On.** The candidate observed the most critical lesson of asynchronous decoupling: *The client experienced zero errors*. The API returned a `202 Accepted`, while the backend silently caught the error, retried 3 times based on the `maxReceiveCount`, and safely routed the bad payload to a Dead Letter Queue (DLQ) without disrupting the parallel "Bike" processing pipeline.

Step 4: Compare 2 Alternatives

- **Candidate's Action:** Evaluated Synchronous and Real-Time Streaming alternatives.
- **Coach's Assessment: Excellent.** The candidate correctly identified the core differences.
 - *Against Sync (API -> Lambda -> DB):* Identified that it lacks fault tolerance and drops orders if the DB throttles.
 - *Against Streaming (Kinesis):* Identified that Kinesis introduces unnecessary cost and shard-management overhead for discrete, independent events.

Step 5: Simulate Pressures (Scale, Failure, Cost)

- **Candidate's Action:** Evaluated Locust for load generation, AWS Free Tier cost implications, and DLQ redrive mechanisms.
- **Coach's Assessment: Solidified.** Recognized that because of the async shock-absorbers (SQS) and on-demand DynamoDB scaling, the architecture easily handles bursts of thousands of requests per second. Acknowledged that Serverless scale-to-zero keeps development costs at \$0.00 under the Free Tier.

Step 6: Write the Observation Sentence

- **Candidate's Action:** Drafted mental models for the architecture's behavior under pressure.
- **Coach's Assessment: Mastered.** Locked in two highly accurate exam shortcuts:
 1. **SNS Message Filtering** behaves as an **intelligent traffic router** when mixed categories enter the fanout topic.
 2. **The SQS Dead Letter Queue** behaves as a **quarantine zone** when poisoned payloads repeatedly crash the downstream Lambda.

Step 7: Map to Exam

- **Candidate's Action:** Extracted IAM permissions and Serverless concepts into exam-ready insights.
- **Coach's Assessment: Spot On.** The candidate successfully identified the "Gotchas" that appear on AWS Professional exams:
 - *IAM Least Privilege:* Lambda requires both an Execution Role to poll SQS, and SQS requires a Resource Policy to allow SNS to publish.
 - *Visibility Timeout Math:* The SQS visibility timeout must exceed the Lambda execution timeout to prevent duplicate processing.

Coach's Final Recommendation

The candidate has successfully proven a deep, hands-on understanding of decoupling, retry mechanisms, fan-out patterns, and Infrastructure as Code. The candidate is operating at the AWS Certified Solutions Architect - Professional level for this domain. Proceed to the next architectural pattern.